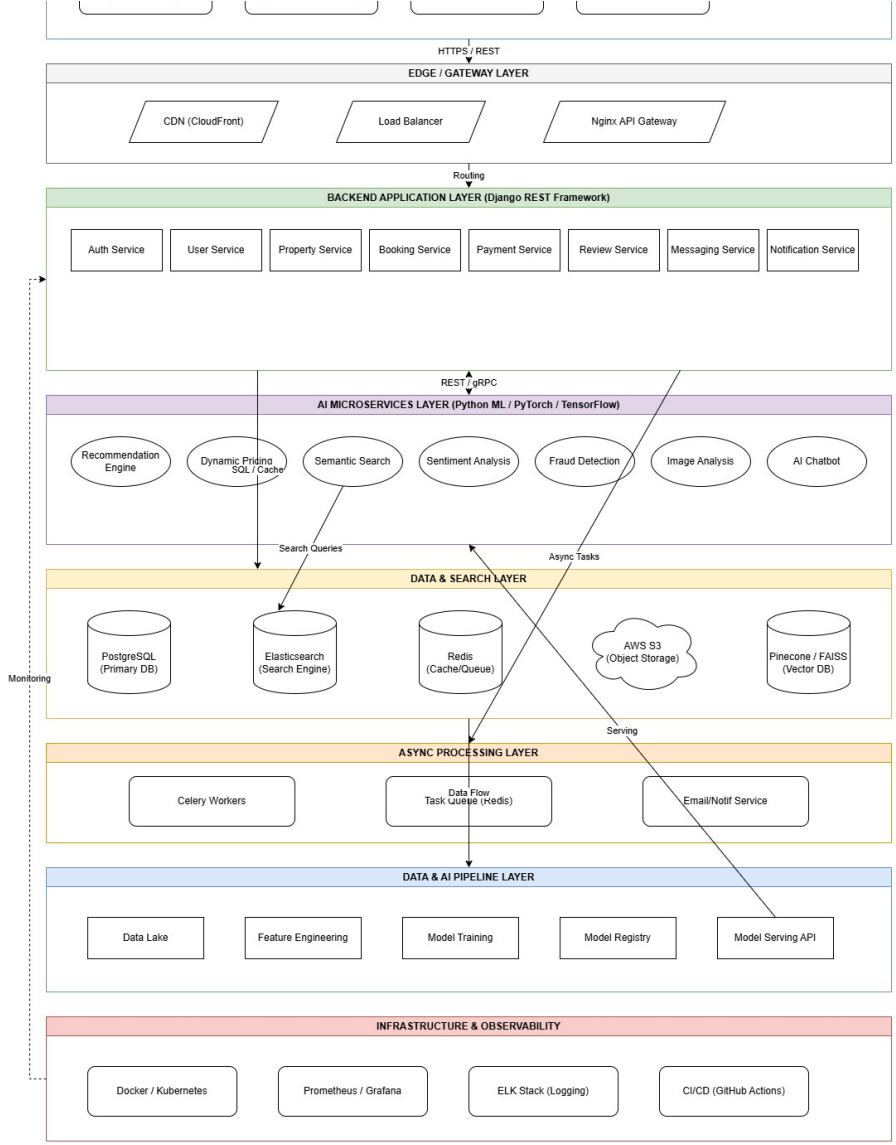
HTTPS / REST
EDGE / GATEWAY LAYER
CDN (CloudFront)
Load Balancer
Nginx API Gateway
Routing
BACKEND APPLICATION LAYER (Django REST Framework)
Auth Service
User Service
Property Service
Booking Service
Payment Service
Review Service
Messaging Service
Notification Service
REST / gRPC
AI MICROSERVICES LAYER (Python ML / PyTorch / TensorFlow)
Recommendation Engine
Dynamic Pricing
Semantic Search
Sentiment Analysis
Fraud Detection
Image Analysis
AI Chatbot
SQL / Cache
Search Queries
Async Tasks
DATA & SEARCH LAYER
PostgreSQL (Primary DB)
Elasticsearch (Search Engine)
Redis (Cache/Queue)
AWS S3 (Object Storage)
Pinecone / FAISS (Vector DB)
Monitoring
Serving
ASYNC PROCESSING LAYER
Celery Workers
Data Flow
Task Queue (Redis)
Email/Notif Service
DATA & AI PIPELINE LAYER
Data Lake
Feature Engineering
Model Training
Model Registry
Model Serving API
INFRASTRUCTURE & OBSERVABILITY
Docker / Kubernetes
Prometheus / Grafana
ELK Stack (Logging)
CI/CD (GitHub Actions)